Listing 10.12 A simple component

```
d3.svg.legend = function() {

    function legend(gSelection) {
      var testData = [1,2,3,4,5];

      gSelection.selectAll("rect")
          .data(testData)
      .enter()
      .append("rect")
      .attr("height", 20)
      .attr("width", 20)
      .attr("x", function (d,i) {return i *25})
      .style("fill", "red")

      return this;
    }

    return legend;
  };
```

A component is sent a selection with .call().

Appends to that selection a set of rectangles

We can then append a <g> element to our chart and call this component, with the results shown in figure 10.8:

```
var newLegend = d3.svg.legend();

d3.select("svg").append("g")
   .attr("id","legend")
   .attr("transform", "translate(50,400)")
   .call(newLegend);
```

And now that we have the structure of our component, we can add functionality to it, such as allowing the user to define a custom size like we did with our grid layout. We also need to think about where this legend is going to get its data. Following the pattern of the axis component, it would make the most sense for the legend to refer directly to the scale we're using and derive, from that scale, the color and values associated with the color of each band in the scale.

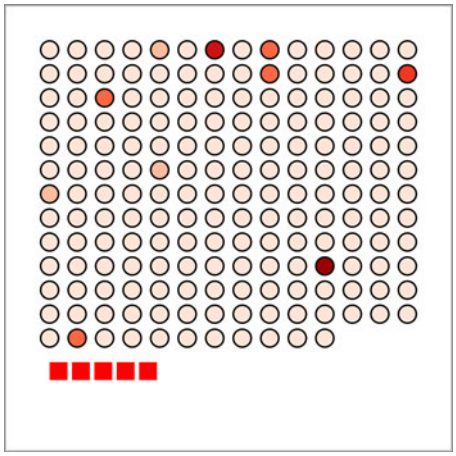


Figure 10.8 The new legend component, when called by a <g> element placed below our grid, creates five red rectangles.